Baylor CSI 3471 Software Engineering I

Stay & Shop Reservation System

Version 3.0
Scrumbags
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1. Introduction

Our project is a Hotel Reservation Website designed to support the core and realistic operations of a modern hotel. The system provides an accessible, user-friendly platform where guests can make reservations; clerks and administrators can manage bookings, hotel operations, and more.

The website has three distinct user roles – guest, clerk, and admin – each with its own login and set of capabilities for the responsibilities of that user type. This structure ensures that every user interacts solely with the features relevant to their needs, which in turn improves usability and security.

The purpose of this system is to streamline the reservation process, reduce the workload for hotel staff, and offer guests a convenient way to interact with the hotel online. The system ultimately demonstrates how software can enhance efficiency and accessibility in the hospitality domain.

The motivation behind this project is to create a functional platform that mirrors real-world hotel management systems while giving users reliable experience. Our goal is to show how thoughtful design and well-structured software can support both customer-facing and internal hotel operations.

2. Project Overview

The Hotel Reservation Website provides a unified platform that supports the essential functions of a real-world hotel environment. From an external perspective, the system enables guests to browse room options, create reservations, and manage their strays, while hotel clerks and admin can oversee bookings, maintain customer records, and manage hotel operations. Each user interacts with the system through role-specific interfaces that present the features relevant to their responsibilities.

Its layout, navigation structure, and labeling push an emphasis for clarity, helping users understand available actions and move through the reservation process with ease. By providing a clean and professional interface, the system reduces confusion and supports task completion. Whether a guest is booking a room or an admin is managing hotel data.

Overall, the system's capabilities reflect the real-world needs of both customers and staff, providing a cohesive digital experience that aligns with the operational expectations of a modern hotel.

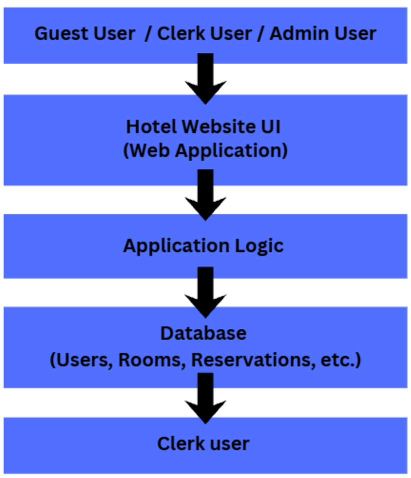
2.1 Product Perspective

The Hotel Reservation Website operates as a standalone web-based system that interacts with a central data base and user-facing interface. From an external perspective, the system consists of three primary sections:

1. The web interface used by guises, clerks, and admins,
2. The application layer that processes user actions, and
3. The database that stores essential hotel information.

The system relies on its database to maintain user accounts, authentication details, reservation records, room availability, and other hotel-related data. This database functions as the system's primary external resource, enabling consistent data access across all user roles. The website interface serves as the main point of interaction, allowing users to perform tasks such as logging in, viewing room options, managing reservations, and accessing admin tools.

With this additional environment, the system acts as a centralized platform that connects users to hotel operations through a structured interface. Its boundaries are clearly defined: the users interact with the website, the website communicated with the database, and all hotel-related information is stored and retrieved through this connection. This context highlights how the system fits into a typical hotel management workflow while remaining accessible.



2.2 Assumptions and Dependencies

The development of this website is based on several assumptions and constraints that shape how the system operates within its environment. These factors influence design decisions, system behavior, and the technologies required for proper functionality.

Assumptions:

- The system is accessed through a modern web browser that supports standard web technologies.
- Users have a stable internet connection when accessing the website.
- All user roles- guest, clerk, and admin-possess valid login credentials stored in the system's database.
- Hotel policies such as room availability, pricing, and reservation rules remain consistent.
- The database contains accurate and up-to-date information, and no external data sources are required for system operation.

Constraints:

- The system relies on a single centralized database; changes to the structure or platform would require corresponding updates to the application.
- Mobile optimization is not within the scope of this project.
- The system does not integrate with external payment processors, meaning all reservation confirmations occur within the system without external financial transactions.

These assumptions and constraints define the boundaries of the system and clarify the conditions the website is expected to function. Any significant change to these factors would require revisiting aspects of the system's design and implementation.

3. Requirements

3.1 Functional Requirements

- Guest User Requirements - the system allows a hotel guest to:
  1. Create a guest account using a unique username and password.
  2. Log in using valid credentials.
  3. Search for available rooms based on date range and room type.
  4. Make a reservation by providing required booking details.
  5. Modify an existing reservation when modification rules allow it.
  6. Cancel a reservation prior to the reservation start date, with cancellation penalties applied according to hotel policy.
  7. Browse store products available during their stay.
  8. Add store items to a shopping cart.
  9. View items in the shopping cart and complete a purchase.
  10. Receive a combined bill that includes both hotel stay charges and store purchases.
- Clerk User Requirements - the system allows a hotel clerk to:
  1. Log in using a valid username and password.
  2. Modify their own profile information, including passwords.
  3. Add new rooms and update room information (type, beds, smoking status, quality level, etc.).
  4. View the status of all rooms (available, reserved, occupied).
  5. Make a reservation on behalf of a guest.
  6. Modify any guest reservation when allowed.
  7. Cancel any reservation prior to the reservation start date, with penalties applied.
  8. Process guest check-in and check-out.

9. Generate billing information for any guest.
10. Add and remove items from the hotel shop.

- Admin User Requirements - the system allows a hotel admin to:

1. Log in using a valid username and password.
2. Create clerk accounts with a username and default password.
3. Reset passwords for any user account (guest or clerk).

3.2 Non-functional Requirements

- Performance
    - The system shall return room search results within 3 seconds under normal load.
    - The system shall support simultaneous access by multiple users without significant performance degradation.
- Usability
    - The system shall provide clear navigation for all user roles.
    - All user actions shall provide confirmation messages (success or failure).
    - Error messages shall be descriptive and guide the user toward corrective action.
- Constraints
    - The system relies on a centralized database; changes to the schema require corresponding application updates.
    - The system is designed primarily for desktop browser use.

3.3 Requirements Clarifications

This subsection documents ambiguities encountered during development and how the team resolved them.

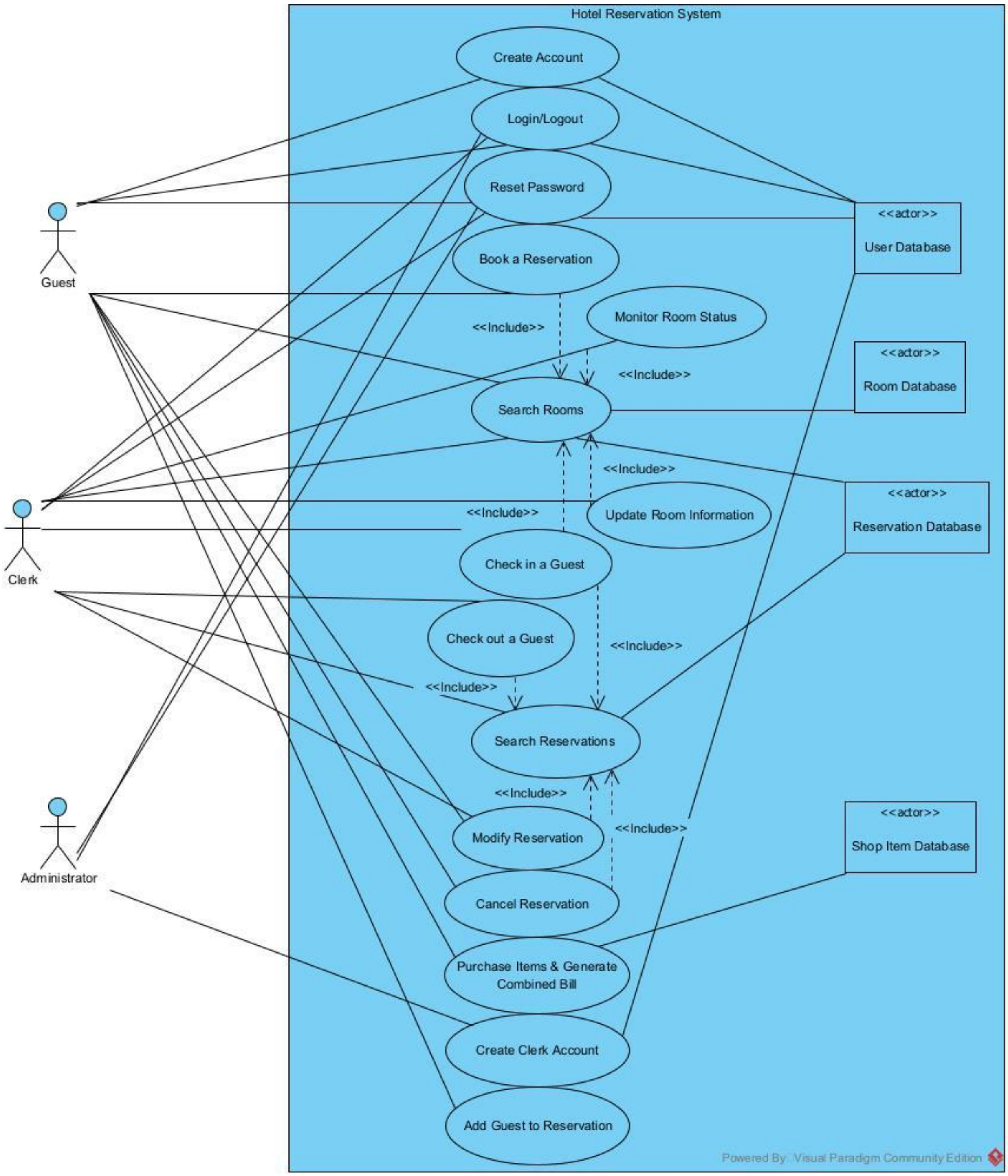
- Clarification 1: Reservation Modification Rules
    - Ambiguity: Can a guest modify a reservation after the start date?
    - Resolution: Guests may modify reservations only before the reservation start date. Clerks may modify reservations at any time if hotel policy allows.
- Clarification 2: Cancellation Penalties
    - Ambiguity: How is the 2-day cancellation window interpreted?
    - Resolution: The system calculates the penalty based on the time elapsed since the reservation was created.

        - Canceled within 2 days → no penalty
        - Canceled after 2 days → 80% of one night's rate charged

- Clarification 3: Room Allocation at Check-In
    - Ambiguity: Does the system assign a specific room at reservation time or check-in?
    - Resolution: Room type is selected during reservation; a specific room number is confirmed at check in.

4. Use Case Model

4.1 Use Case Diagram



Author: Matt Freeman

4.2 Fully-Dressed Use Cases

USE CASE NAME	***CHECK IN A GUEST***
Author	*Matt Freeman*
Actor	*Clerk*
Preconditions	*The reservation exists in the system and starts today*
Postconditions	*The reservation is set to active*
Main Success Scenario	*1. A guest approaches the clerk to check in.* *2. The clerk searches for the guest's reservation by name or confirmation number.* *3. The system displays the reservation and room specifications.* *4. The clerk selects "Check-In."* *5. The system sets reservation to active and returns the room number*
Extensions	***1. The reservation doesn't start today (step 4)*** ***a.*** *The system asks the clerk to confirm checking in the guest on the wrong day* ***2. The reservation doesn't exist in the system (step 2)*** ***a.*** *The system will display no search results*

USE CASE NAME	***CREATE ACCOUNT***
Author	*Andrew Hutcheson*
Actor	*Hotel Guest*
Preconditions	*1. Guest does not already have an account associated with their email.* *2. The login system works and is accessible.*
Postconditions	*1. A new guest account is created for the user's email and stored in the system.* *2. The guest can successfully login with their new credentials.*
Main Success Scenario	*1. The guest initiates account creation.* *2. The guest enters their email, password, and any other required credentials.* *3. The guest submits the form.* *4. The system checks whether an account already exists for the provided email.* *5. The system creates and saves the guest account.*
Extensions	*1. Invalid Input:* *If the guest's email is already associated with an account or if other required information is invalid, the system prompts the guest to sign in or enter a different email address.*

USE CASE NAME	***CREATE CLERK ACCOUNT***
Author	*Emmanuel Humber*
Actor	*Admin*
Preconditions	*1.Clerk username is unique* *2. Login system is up accessible*
Postconditions	*1. A new clerk account with the associated Employee ID will be created.* *2. Clerk will have an individualized clerk interface*
Main Success Scenario	*1. Admin selects create account button* *2. Admin enters username, password, and Employee ID prompted by system.* *3. A new clerk account has been created.* *4. System validates and stores clerk account information*
Extensions	***Account without unique username*** *a. System detects accounts with a matching username.* *b. Prompts Admin with username already taken error message.* ***System Error*** *a. If the system is down, it will prompt system is currently down error message.*

USE CASE NAME	***BOOK RESERVATION***
Author	*Hannah Ross*
Actor	*Guest*
Preconditions	*1. Guest is logged in.* *2. The system shows available rooms.*
Postconditions	*1. A reservation is saved.* *2. Room is reserved for specific dates.*
Main Success Scenario	*1. Guest selects desired room.* *2. System requests Guest info: name, address, credit card, stay dates.* *3. Guest enters required details.* *4. System validates credit cards, addresses, and availability.* *5. System calculates stay cost and provides summary of information.* *6. Guest approves cost amount and details of the reservation.* *7. System saves the reservation.* *8. System displays confirmation number.*
Extensions	***1. Invalid Search Criteria (Step 4)*** *a. The system detects invalid input (e.g. check-out date earlier than check-in date).* *b. The system displays an error message and requests corrected input.*

USE CASE NAME	*CANCEL RESERVATION*
Author	*Duane Schlottke*
Actor	*Guest / Clerk / Admin*
Preconditions	*The reservation exists in the system*
Postconditions	*The reservation is removed in the system and Guest's account*
Main Success Scenario	*1. A guest wishes to cancel their reservation* *2. The guest logs into their account* *3. The system displays active reservations* *4. The guest selects their reservation and selects cancel reservation* *5. The system prompts the user to confirm* *6. The system sets the reservation to inactive and removes it from the User's account*
Extensions	*1. The reservation is occurring in <2 days* *a. The system will prompt the guest to confirm with the added knowledge that there will be a penalty* *b. The system removes the reservation upon confirmation*

4.3 Casual Use Cases

USE CASE NAME – SEARCH RESERVATIONS
Author: Matt Freeman
Main Success Scenario: The clerk searches for existing reservations by reservation ID or guest's name. The system returns all reservations matching the input search criteria.
Alternate Scenarios: If no matching reservations exist, then the system will display no results.

USE CASE NAME – LOGIN
Author: Andrew Hutcheson
Main Success Scenario: The user enters the system's login page and provides their username and password. The system verifies the credentials against stored account data, determines the user's role (guest, staff member, or admin), and authenticates the user. Upon successful authentication, the system grants access and directs the user to the appropriate dashboard based on their role.
Alternate Scenarios (if applicable): If the user enters incorrect credentials, the system displays an error message and allows the user to retry logging in. After multiple failed attempts, the system temporarily blocks further login attempts.

USE CASE NAME – CREATE CLERK ACCOUNT
Author: Emmanuel Humber
Main Success Scenario: The admin selects create Clerk Account and is directed to an account creation tab. The admin then enters the clerk's respective employee ID, username, password, and full name. The admin selects the Register Clerk button, and the system then validates the Clerks username in the database. Once validation is passed, the Clerks account information is stored, and the Clerk can log in with his new credentials.
Alternate Scenarios (if applicable): If Admin enters a username that's already taken in the database, the system will display an error message informing the admin that the username is already taken.

USE CASE NAME – ADD ROOM
Author: Hannah Ross
Main Success Scenario: The Clerk logs into the system and provides the required information to create a hotel room, including room number, floor type, smoking status, rate, and availability. The system verifies the data, processes the request, and confirms the addition to the hotel system.
Alternate Scenarios (if applicable): If the clerk cannot login, the system provides an option for administrators to create another account for their use. If the system is down, the clerk is unable to login or view any options to add a room. When incorrect or inconsistent information is entered (e.g. price updates or inputting an incorrect room number), the system prompts the clerk to correct and re-enter the necessary details.

USE CASE NAME – ADD GUESTS
Author: Duane Schlottke
Main Success Scenario: The clerk/guest wishes to add a guest to the system. The system prompts for information necessary (Name, email, DOB, etc.) and if all are valid, will add new guest.
Alternate Scenarios (if applicable): If guest with credentials already exists, system will display error.

4.4 Brief Use Cases

USE CASE NAME – Check out a Guest (Matt Freeman): When a guest currently checked into the hotel desires to check out, they can do so with the help of the clerk. The clerk will search for their reservation by name or ID, and they will select "Check out." At this point, the reservation will be set to inactive in the system, and the guest's combined bill for the stay will be shown.

USE CASE NAME – Purchase Items & Generate Combined Bill (Andrew Hutcheson): When a guest who is currently staying at the hotel wants to purchase items from the hotel store, they can go to the shopping cart, select one or more available items, and submit the purchase. The system verifies the request, checks item availability, and processes the purchase by logging the items as bought under the guest's account. The system then calculates the total cost of the purchased items and updates the guest's hotel bill by adding these charges to the room fees. Finally, the system generates an updated combined bill that reflects both room costs and any store purchases made during the stay.

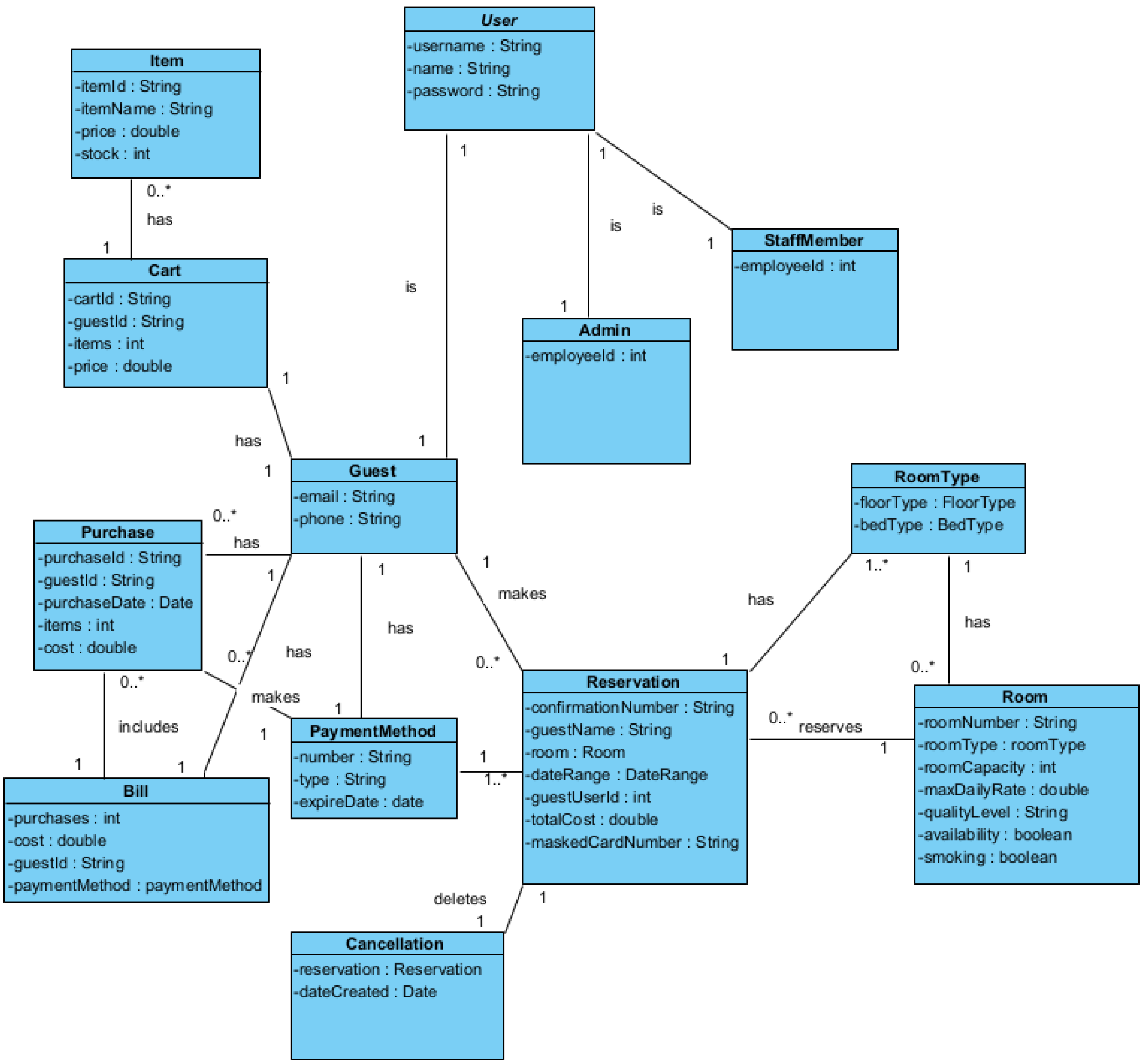
USE CASE NAME – Log Out (Emmanuel Humber): When a guest who is actively in a user session wants to log out, they can press the log out button in the top left corner of the welcome tab, where they will be redirected to the log in screen, and their user session will be ended and any data or information made during their session will be saved to the database.

USE CASE NAME – Reset Password (Emmanuel Humber): When a guest with a valid account in the database forgets their password or wants to change it, they may press the reset password button on the log in tab. This will redirect to another tab prompting input for their account email. Once inputted, if the email corresponds to an account in the database, they will be redirected to another screen to input and confirm their new password. If the email does not correspond, they will receive an error message saying no email in the system.

USE CASE NAME – Manage Store Products (Hannah Ross): A clerk can sign into the System and perform many actions regarding the store for the hotel. These actions include involving inputs for price and quantity for existing products, as well as adding products of interest. To add an item, the System requires a SKU, name, optional description, and price.

USE CASE NAME – Cancel Reservation (Duane Schlottke): If a guest so chooses to remove their reservation, they have the option to cancel it. The guest or clerk can cancel the reservation, which if cancelled more than 2 days in advance will be penalty free. If cancelled less than 2 days in advance, there will be a monetary penalty.

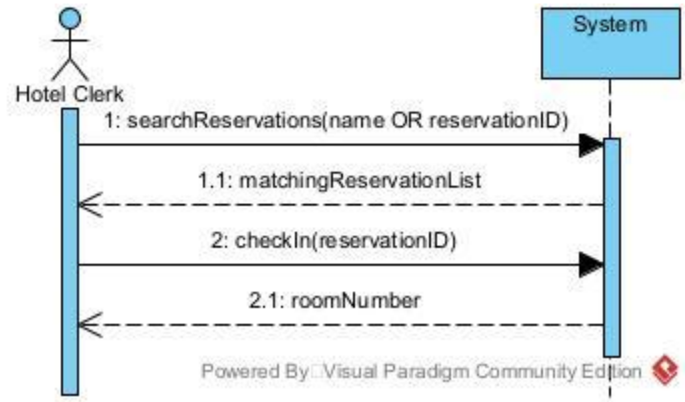
5. Domain Model (Analysis)



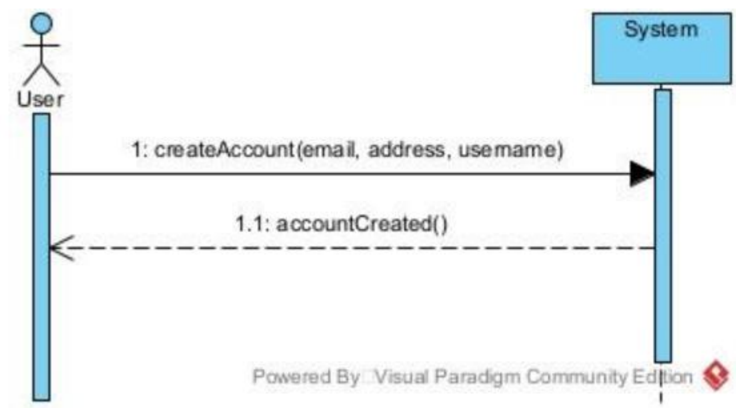
Author: Emmanuel Humber

6. System Sequence Diagrams (SSD)

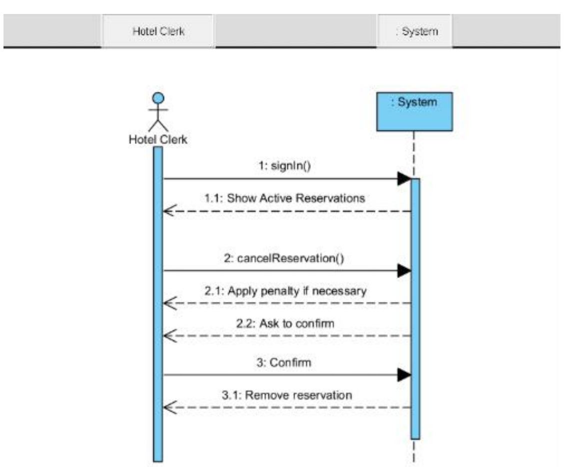
USE CASE NAME – Check in a Guest (Matt Freeman):



USE CASE NAME – Create Account (Andrew Hutcheson):



USE CASE NAME – Cancel Reservation (Duane Schlottke):



7. Operation Contracts

Contract CO1: CHECK IN A GUEST

Author: Matt Freeman

***Operation**: checkIn(reservationID)*

Cross References: Use Case – Check in a guest

Preconditions:

1. ***Clerk is logged in***
2. ***Reservation exists in the system and starts today***

Postconditions (Main Success):

1. ***Reservation is set to active (guest will be able to shop)***
2. ***Room number is returned and set to occupied***

Contract CO1: Create Account

Author: Andrew Hutcheson

***Operation**: createAccount()*

Cross References: Use Case – createAccount

Preconditions:

1. ***User does not have an existing account***

Postconditions (Main Success):

1. ***User account is successfully created and stored by system***
2. ***User gets confirmation of account being created***

Contract CO2: CANCEL RESERVATION

***Author:** Duane Schlottke*

***Operation**: cancelReservation()*

Cross References: Use Case – Cancel Reservation

Preconditions:

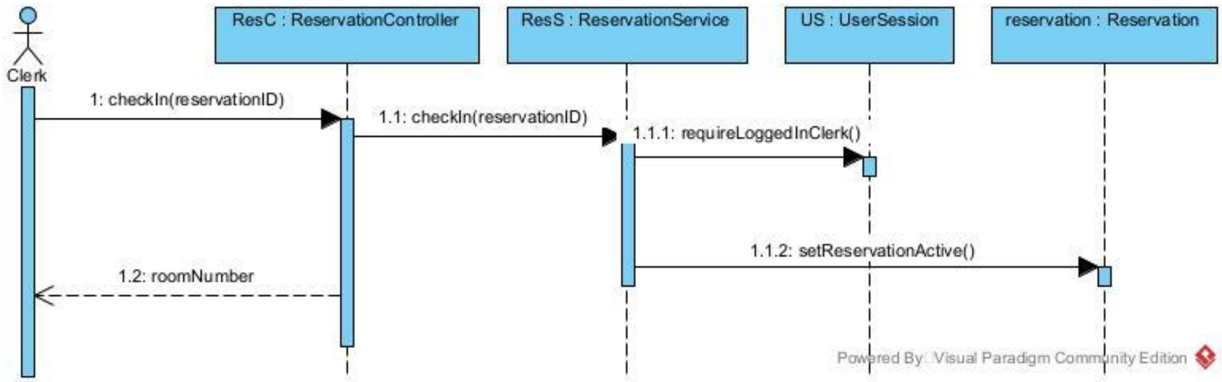
1. ***Guest/Clerk/Admin is logged in***
2. ***Reservation exists in the system***

Postconditions (Main Success):

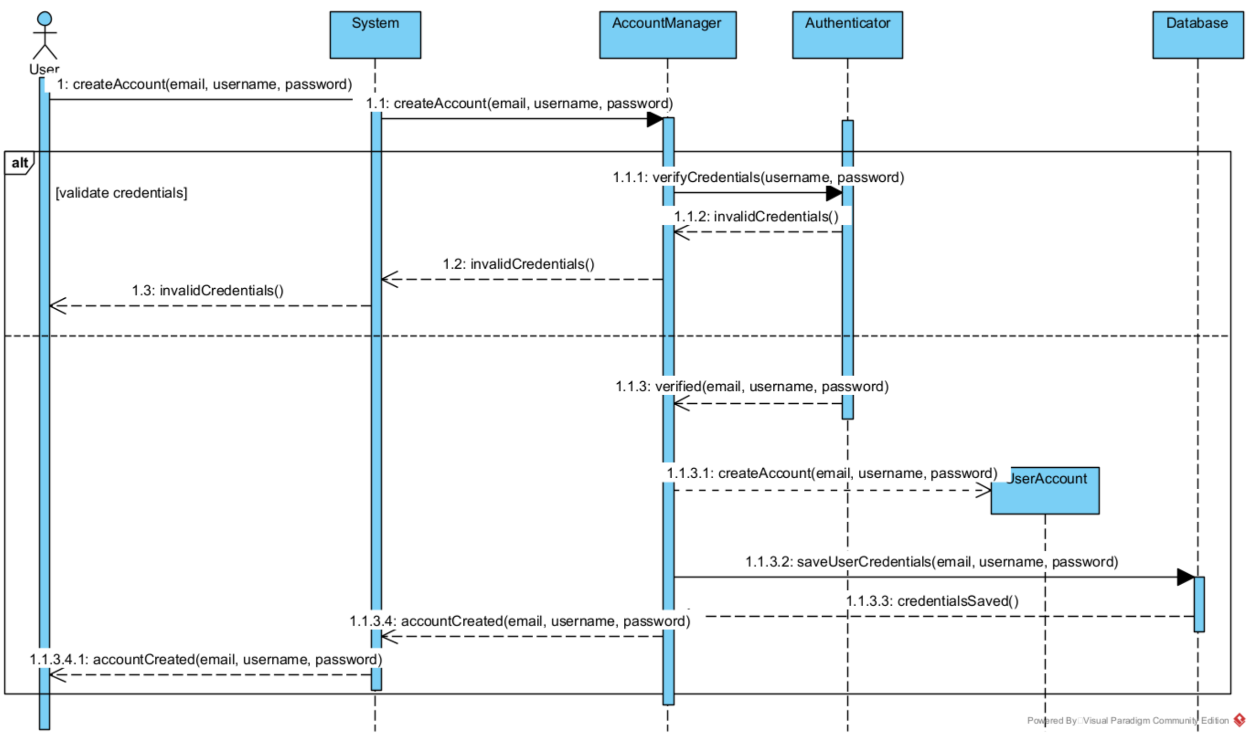
1. ***Reservation is set to inactive***
2. ***Reservation is removed from guest's account***

8. Sequence Diagrams (Design)

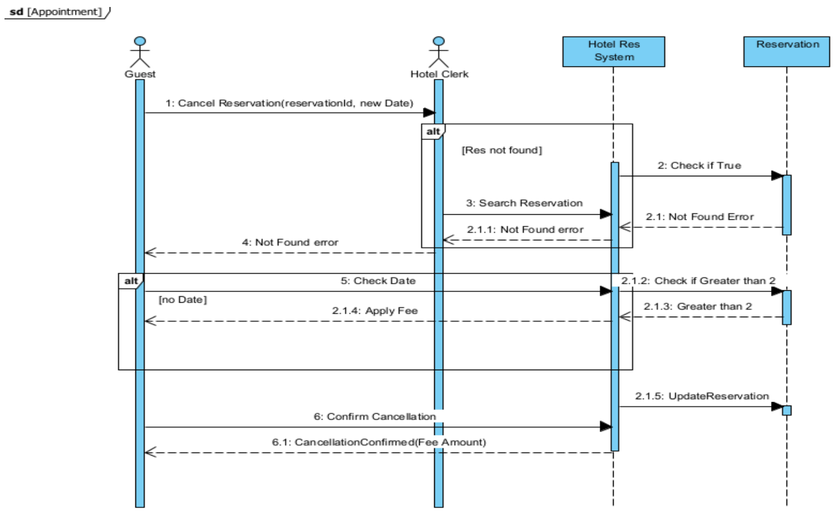
USE CASE NAME – Check in a guest (Matt Freeman):



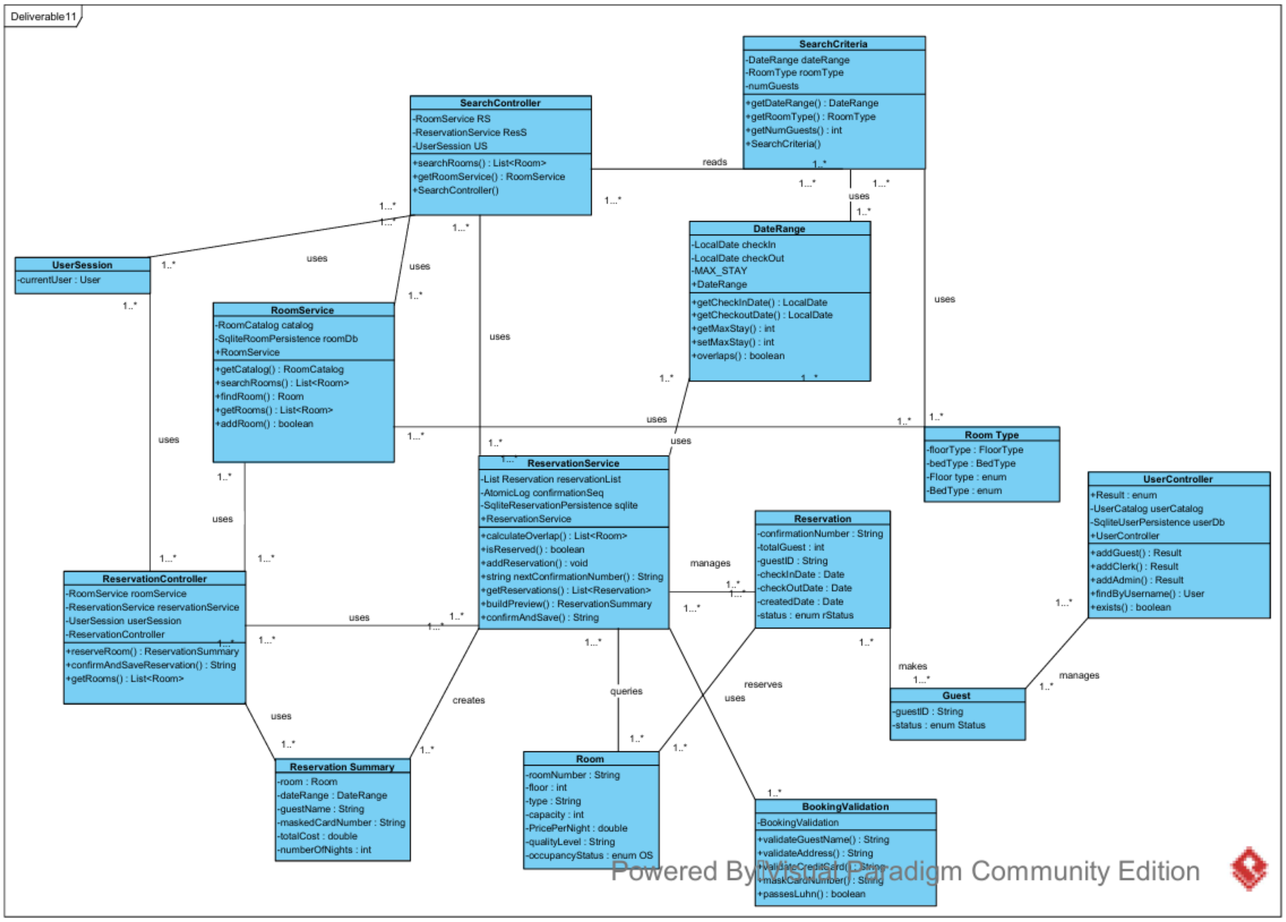
USE CASE NAME – Create Account (Andrew Hutcheson):



USE CASE NAME – Cancel Reservation (Duane Schlottke):



9. Design Class Diagram (DCD)



Author: Hannah Ross

10. Implementation Overview

10.1 Architecture

The structure of the system includes five main layers of function: UI, Controllers, Services, Domain, and Technical Services. The UI layer is what the user sees and interacts with, which provides them with a framework for using the functionality of the controllers. The Controllers layer handles system operations like checkIn() and searchRooms() and calls on the services to do work. The Services layer has direct interaction with the domain objects to create, delete, and modify them while storing them in the database. The Domain layer houses all of the real world concepts like Room, Reservation, Clerk, etc., with each knowing how to perform operations specific to itself. Finally, the Technical Services layer includes our database for persisting information as well as some security features.

10.2 Key Design Decisions

We kept a UI -> Controller -> Service -> Domain and technical services flow even when some controller methods only forward to a service. That adds a little boilerplate but keeps every feature in the same structure, so screens stay free of business rules and SQL and the code strucutre is consistent and easy to follow. Driver.Driver acts as the composition root: it builds services, SQLite persistence, controllers, a shared UserSession, and MasterUI against one database path. Central wiring avoids duplicate connections and mismatched configuration across guest, clerk, and admin experiences.

UserSession holds the logged-in User (guest, clerk, or admin via subclasses) and can store a PendingReservation so a flow that starts on the public shell can continue after login. MasterUI uses that state to switch CardLayout shells and refresh panels. Services are the information experts: ReservationService owns reservation rules, ShoppingService owns cart and checkout (including the rule that shopping requires an active stay), and UserService owns accounts. Controllers stay thin and expose coarse operations the Swing layer can call without touching JDBC.

Persistence lives under TechnicalServices/Persistence, with schema from schema.sql applied through SchemaInstaller, so domain types do not carry JDBC. Reservation charges are stored at confirm time; store purchases are separate rows. buildCombinedBill aggregates both at read time into CombinedBill for the UI, which gives one total to the guest without rewriting reservation history when someone shops. PasswordHasher concentrates password handling so verification and upgrades stay in one place. Together, these choices keep the desktop app consistent, testable, and aligned with separation of concerns for a multi-role hotel and store system.

11. Testing

11.1 JUnit Testing Summary

Our test plan has been, more or less, that we test as we go. Whenever one of us implemented a new feature, whether initially as only a backend functionality or later as a fully-fledged use case with UI connectivity, that team member would test it and confirm that it worked as intended in all respects before moving on to the next feature. To know how to test the system and what kinds of inputs and edge cases to try, we referred to our use cases as well as our experience with other similar hotel systems.

11.2 Test Cases Table

Test Case ID	***#1***
Assigned Tester	*Hannah*
Scenario/Condition	*Verify that a clerk can successfully login and access the make reservation functionality.*
Test Inputs	*1. Username: "clerk"* *2. Password: "clerk123"*
Expected Result	*1. System logs in the clerk user.* *2. Screen displays clerk-only options, including add room and make reservations.*
Actual Result	*1. System correctly logs in the clerk.* *2. Dashboard displays add room and make reservation buttons as expected*
Test Status	***Passed*** *or Failed*

Test Case ID	***#2***
Assigned Tester	*Hannah*
Scenario/Condition	*Verify that the clerk can click the add reservation button, and the screen displays the interactive window to make a reservation.*
Test Inputs	*1. Username: "clerk"* *2. Password: "clerk123"*
Expected Result	*1. System logs in clerk user.* *2. Clicking on the "reservation" button opens a separate window with the various options to make a reservation. Should display:* *a. List of all reservations.* *b. Input section (Confirmation number, Room, Check-in and Check-out date, Guest name, Credit card number, and optional Guest username).* *c. Buttons (Refresh list, New (Clear form), Calculate cost, Confirm new reservation, Save changes, Delete selected).*
Actual Result	*1. System correctly logs in the clerk* *2. After clicking "Reservations", the System displays the window composed of all expected components listed above.*
Test Status	***Passed*** *or Failed*

Test Case ID	*#3*
Assigned Tester	*Emmaunel Humber*
Scenario/Condition	*Verify that system correctly validates Clerk information when a new clerk account is created. System should throw error messages for non-unique ID's and usernames.*

Test Inputs	`1.addClerk(1, "sameuser", "pw1", "First")`
	`addClerk(2, "sameuser", "pw2", "Second")`
	`2.addClerk(99, "clerkA", "pw1", "A")`
	`addClerk(99, "clerkB", "pw2", "B")`

Expected Result	*1. System should prompt Admin with duplicate username message.* *2. System should prompt Admin with duplicate Employee ID message*
Actual Result	*1. System successfully displayed duplicate username error message to admin.* *2.System successfully displayed a duplicate Employee ID error message to Admin.*
Test Status	*Passed*

Test Case ID	*#4*
Assigned Tester	*Emmaunel Humber*
Scenario/Condition	*Verify that Clerk Account information is saved to database, and that clerk can succesfully log-in and view their user interface*
Test Inputs	`1.addClerk(42, "newclerk", "SecurePass1", "Pat Clerk")` `login("newclerk", "SecurePass1")`
Expected Result	*1. Clerk should be able to log-in to their own respective user interface.*
Actual Result	*1. Clerk log-in and view of their user interface was successful.*
Test Status	*Passed*

Test Case ID	*#5*
Assigned Tester	*Matt Freeman*
Scenario/Condition	*Verify that a logged in guest on the "Make Reservation" page, having selected room available for certain dates, can enter information, calculate cost, and confirm reservation as expected.*
Test Inputs	*1. Enter "Test Guest" as guest name* *2. Enter 1234123412341234 as credit card number* *3. Click "Calculate Cost"* *4. Click "OK" when summary is displayed* *5. Click "Confirm Reservation"* *6. Click "OK" to end*
Expected Result	*1. Guest can see Reservations page and enter all information, which will be displayed after being entered* *2. When guest clicks "Calculate Cost", a reservation summary is shown with relevant information* *3. When guest clicks "Confirm Reservation", a confirmation number will be shown and the reservation will be added to the system*
Actual Result	*1. System correctly displays Reservations page and allows the guest to enter information* *2. The correct reservation summary is shown when "Calculate Cost" is clicked* *3. A confirmation number is displayed when "Confirm Reservation" is clicked*

	4. The reservation is saved to the database, which can be confirmed by checking the clerk's reservation view or using a database viewer
Test Status	*Passed*

Test Case ID	***#6***
Assigned Tester	*Duane Schlottke*
Scenario/Condition	*Verify that a logged in guest on the account information page can enter the Shopping Cart, as well as view items in the database.*
Test Inputs	1. *Enter "Test Guest" as guest name* 2. *Enter "1234123412341234" as credit card number* 3. *Click "Shopping Cart"*
Expected Result	1. *Guest should be able to see items in their shopping cart* 2. *Guest should be able to see items available for sale in database*
Actual Result	1. *System correctly opens shopping cart* 2. *System correctly displays items for sale from System database, Guest is able to scroll and view items in Shopping Cart as well as items for sale*
Test Status	*Passed*

TEST CASE ID	***#7***
Assigned Tester	*Andrew*
Scenario/Condition	*Verify that a logged-in guest on the make-reservation flow (after a room and stay dates are selected, same as Calculate Cost in ReserveRoomUI) can request a cost preview only and receive a summary with the correct room, stay length, total, and guest name—without confirming a reservation.*
Test Inputs	*Sign in as a guest account whose profile name is "Test Guest" (for automation: username g3, password guestpw1).* *Assume room 303 is available and priced at $100.00 per night.* *Check-in date: today + 5 days.* *Check-out date: today + 8 days .* *Guest name field: "Test Guest".* *Credit card: "4111111111111111".* *Click "Calculate Cost"*
Expected Result	*The system shows a reservation summary (dialog title "Reservation Summary" in the UI).* *The summary corresponds to room 303, guest name "Test Guest", 3 nights, and total cost $300.00 (3 × $100.00).*
Actual Result	*The automated test completed successfully. After signing in as guest g3, calling Calculate Cost , the system produced a preview with room 303, 3 nights, total $300.00, and guest name "Test Guest".*
Test Status	*Passed or Failed*

Test Case ID	*#8*
Assigned Tester	*Andrew*
Scenario/Condition	*Verify that when a logged-in guest clicks Calculate Cost with an invalid credit card number, the system rejects the request and does not create a reservation.*
Test Inputs	*Sign in as a guest whose profile name is "Test Guest" (username g4, password guestpw1).* *Room 404 available ($80.00/night for reference only).* *Check-in: today + 7 days; check-out: today + 9 days.* *Guest name: "Test Guest".* *Credit card: "123" (too few digits).* *click "Calculate Cost".*
Expected Result	*The system does not show a valid reservation summary.* *The user sees an error indicating the credit card is invalid (e.g. message that the number must be between 13 and 19 digits).* *No reservation is stored.*
Actual Result	*The automated test completed successfully. With guest g4 signed in, attempting Calculate Cost for room 404 with credit card "123" caused an IllegalArgumentException with message "Credit card number must be between 13 and 19 digits."*
Test Status	*<u>Passed</u> or Failed*

12. Team Contribution

- Hannah
    - Estimated Contribution Percentage: 25%
    - Design Engineer
    - Use Cases: Book a Reservation, Update Room Information, Monitor Room Status
    - Responsible for bridging the gap between user experience, user interface, and technical implementation.
    - I contributed to designing the overall UI and flow of the website.
    - I also developed my use cases and several of the clerk-side functionalities.
    - Towards the end of each iteration, I conducted structured testing to ensure the website functioned as intended and met the team's expectations for usability and performance.
    - Regarding planning, I created the Design Class Diagram.
- Andrew
    - Estimated Contribution Percentage: 25%
    - Requirements Engineer
    - Use Cases: Login, Create Account, and Purchase Items & Generate Combined Bill
    - Responsible for implementing the backend login connected services to database.
    - I contributed to designing the database structure and ensuring we stored the correct data and could retrieve and update it efficiently.
    - Worked on search functionalities in guest, clerk, and admin whenever necessary, such as Admin reset password, and clerk modify room information.
    - Towards the end of each iteration, I collaborated with Matt to make sure we had all the features we wanted and that there were no last-minute fixes needed for our current features.
    - Helped write tests using junit to ensure that our code was properly tested.
- Matt
    - Estimated Contribution Percentage: 25%
    - Librarian
    - Use Cases: Check In, Check Out, Search Reservations
    - Responsible for keeping code well-structured throughout development, refactoring code into different classes to maintain consistency and GRASP principles when necessary
    - I set up the initial structure of our code and coded significant portions of several of our use cases such as Modify/Cancel a Reservation, Edit Profile, Check In/Out, Monitor Room Status, etc.
    - I kept things well-documented and created templates for most team deliverables as well as communicating with the team about what worked remained to be done prior to major deadlines
- Emmanuel
    - Estimated Contribution Percentage: 15%
    - Quality Assurance Engineer
    - Use Cases: Log Out, Create Clerk Account, Reset Password

  - Created Calendar Window UI for Reserve Room Use Case.
  - Assisted in Reserve Room Overlap Validation as well as initial Reserve Room implementation testing.
  - Responsible for original design of project through creating conceptual ideas in Domain Model as well as their attributes.
  - Assisted with quality-of-life updates and logical errors mainly with booking reservations and rooms.
  - Created the testing package and tested my use cases with JUnit implementation as well as generic testing with overall project.
- Duane
  - Estimated Contribution Percentage: 10%
  - Project Manager
  - Use Cases: Cancel Reservation, Add Guests, Shopping Cart Functionalities
  - Responsible for creating preliminary home page, shopping cart page, and basic reservation functions
  - Contributed to overall UI
  - Developed Use Cases and Test Cases for shopping cart functions, certain reservation functions
  - Created slideshow for iteration 1
  - Created diagrams for use cases
  - Responsible for creating initial workflow schedule, Trello, etc.

13. Setup / Installation Guide

Required tools

Java Development Kit (JDK) 25 (matches the compiler release in Implementation/pom.xml).

Apache Maven (3.6 or newer recommended), available on your system PATH.

Environment

No extra environment variables are required. The app uses an embedded SQLite database.

The database file is created at data/database.db relative to the directory you run Maven from (normally SE-Group-Project/Implementation).

Note: Admin account is username: "admin", password: "admin123". With this, any new clerk or guest accounts can be made

Steps

1. *Install JDK 25 and Maven, and confirm they work:*

   Java -version
   mvn -version

2. *Open a terminal and change to the Maven project root:*

   SE-Group-Project/Implementation
   (the folder that contains pom.xml).

3. *Build and launch the desktop application:*

   mvn

   The project's default Maven goal compiles the code and runs Driver.Driver (the Swing UI).

4. *Optional: run automated tests:*

   mvn test

5. *Resetting data*

   To start with an empty database, delete data/database.db or the entire data folder under Implementation, then run mvn again.

14. User Manual

All users homepage:

- To Search Rooms:
  - Enter various criteria for check-in date, check-out date, smoking status, floor type, and bed type on the home page, then hit search rooms
  - A list of available rooms will be displayed, and the user can choose a room, then again choose dates (to confirm previous choice) on a calendar, and they will be prompted to sign in as a guest to complete the reservation
- To Login (admin, clerk, or guest):
  - Click "Login" on the navigation bar, then enter username and password and click "Login" below the text fields
- To Create a Guest Account:
  - From the Login page, click "Create account", then enter required information and click "Register User"
  - Now this created account can be logged into
- To Reset Password if Forgotten (as guest):
  - Click "Login" then "Reset guest password"
  - Enter your email to be able to reset password

Guest homepage:

- To Make a Reservation:
  - Click on "Search & Book Rooms" and search available rooms as described above
  - The guest will then be redirected to a reservation screen from which they enter their information, click "Calculate Cost" to view a reservation summary, and click "Confirm Reservation" to confirm it
- To Shop (only available while checked in):
  - Click "Shopping" to go to the page, then for as many items as desired, select the item, enter a quantity in the bottom left corner, and click "Add to cart"
  - When all items have been added, click "View cart" (on the shopping page or the homepage)
  - From here, item quantities can be changed just as above, or removed using the buttons on the bottom
  - When done modifying the order, click "Checkout / Purchase" (or "Combined Bill" from the homepage) to see a combined bill.
    - This would actually be paid with a card on file most likely, but this isn't fully implemented since users of a fake system don't actually pay
- To Modify or Cancel a Reservation:
  - NOTE: Cancelling a reservation or modifying its itinerary (dates and/or room) more than two days after the reservation was created results in a fee of 80% of the single night rate for the room. The system will inform you of the charge before you confirm.
  - Click "Manage Reservations" or click on the reservation in the "Your reservations" window
  - From here, click the desired reservation on the left, then click "Cancel Selected" to cancel it, "Modify Itinerary" to modify dates and/or room, or "Update Details" to change non-crucial details like name and card number
  - When Modifying Itinerary or Updating Details, enter the requested information, searching for available rooms if needed, and click okay

- To Edit Profile:
  - Click "Edit Profile" and enter updated information and click "Save Changes"
  - Leave password fields blank if not editing password
  - To edit password (if known), also fill password fields
  - If you forgot your password, an admin must reset it for you, or you can reset it with email as described above
- To Logout:
  - Click "Logout" in the top right corner, and click "OK"

Clerk homepage:

- To Add a Room to the Hotel System:
  - Click "Add Room" and enter all requested information, then click "Save room to catalog"
- To Modify an Existing Room:
  - Click "Modify room information"
  - Search for the room by number, or by scrolling, select the desired room, then enter all update information and click "Save room changes"
- To Check In or Check Out a Guest:
  - Click "Check in / Check out guest"
  - Find the desired reservation by scrolling or by filtering by guest name or confirmation number
  - Click on the desired reservation, then click "Check In" or "Check Out", confirming when prompted
  - Checking out a guest will also display the guest's combined bill for the reservation
  - A checked in reservation will display on this page as "ACTIVE"
- To Create, Modify, or Delete a Reservation:
  - Click "Reservations" to enter the reservations page
  - Create:
    - Click "New (clear form)", then enter requested information, click "Calculate cost", then click "Confirm new reservation"
  - Modify:
    - Select desired reservation from above, searching by name or confirmation number if needed
    - Modify information shown below, then click "Save changes"
  - Delete:
    - Select desired reservation from above, searching by name or confirmation number if needed
    - Click "Delete selected" and confirm
- To View a Guest's Bill:
  - Click "Reservations"
  - Select desired reservation from above, searching by name or confirmation number if needed
  - Click "View guest bill"
- To Monitor Room Status:
  - Click "Monitor room status"
  - From this page, much information about each room can be seen
  - A room's status will show as:
    - "OCCUPIED" if a reservation is currently checked into this room. The reservation's info will be shown on the right in this case
    - "RESERVED" if a reservation is not currently checked into the room, but a reservation for that room starts today. This reservation's info will be shown on the right in this case

        - "UNAVAILABLE" if the room is not occupied or reserved but otherwise unavailable (e.g. for maintenance)
        - "AVAILABLE" if the room is not currently occupied and no reservation starts in it today. The next date the room is reserved for (if there is one) is shown on the right in this case
- To Edit or Add Store Products:
    - Click "Store products" to enter products page
    - Edit:
        - Click desired product in the window to edit
        - Change quantities in selected product window
        - Click "Save price & stock" to save
    - Add Product
        - Navigate to "Add new product" section
        - Type a SKU in the textbox
        - Type a Name in the textbox
        - Type a Description in the textbox (if desired)
        - Type Unit price in textbox
        - Type Initial stock in textbox
        - Click "Add product" to save
- To Edit Profile or Logout:
    - See Guest instructions above (the process is identical)

Admin homepage:

- To Create a New Clerk Account:
    - Click "Create clerk account"
    - From here, enter all requested information (including a unique employee ID), and click "Register Clerk"
    - This new clerk account can now be logged into
- To Reset User Password:
    - Click "Reset user password"
    - Select the username of the account for which to reset the password, searching or scrolling when needed
    - Enter the new password twice, and click "OK"
    - This user will now be able to login under the new password
- To Edit Profile or Logout:
    - See Guest instructions above (the process is identical)